

Session Objectives

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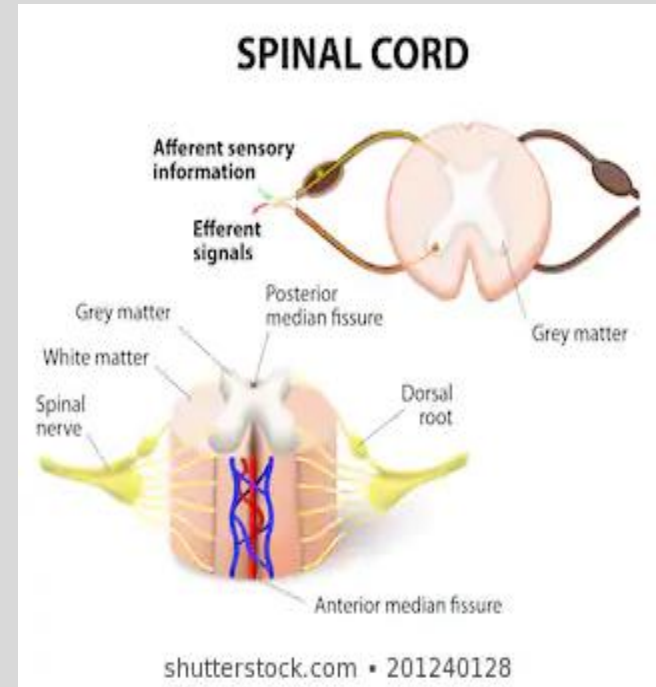
- Objective 1-**Describe** the ascending and descending spinal cord tracts.
- Objective 2-**Explain** common spinal cord syndromes based on cord location.
- Objective 3- **Generate** a differential diagnosis based on the location of the spinal cord injury.
- Objective 4- **Apply** the proper work up for a spinal cord lesion.
- Objective 5- **Recognize** a neurosurgical emergency.

A Review of Cord Anatomy

- Cord ends at L1
- Central grey matter-
 - Contains neuronal cell bodies and synapses.
- Peripheral white matter.
 - Contains ascending and descending fiber pathways.

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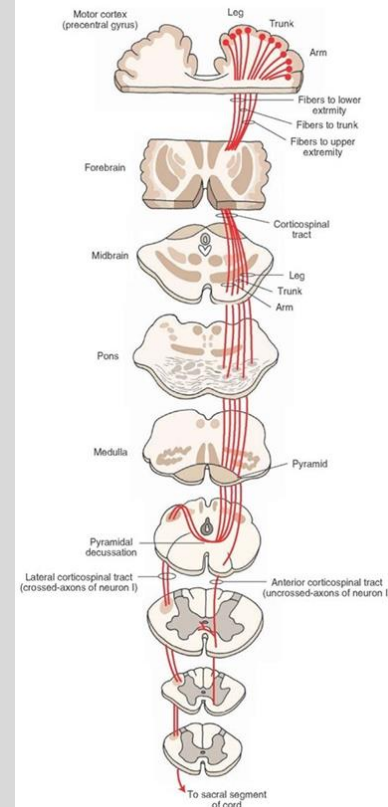
Descending Tract

- **Corticospinal** tract- begins in primary motor cortex.
- Long tract- primary motor cortex in the precentral gyrus to the anterior horn.
- At the level of the medulla the majority cross over and become the lateral corticospinal tracts.

Localization in Clinical Neurology, Wolters Kluwer, eighth edition, 2022

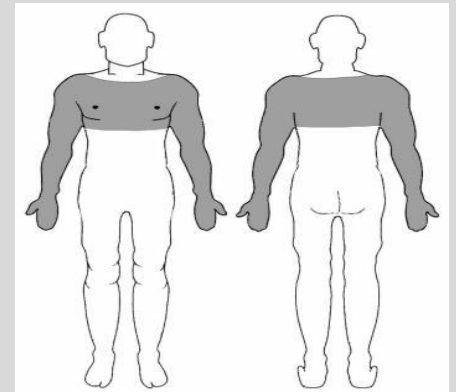
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Case 1

- A 30-year-old otherwise healthy woman presents to an outpatient clinic complaining of **loss of pain and temperature in a capelike distribution.**
- Sensory and motor are also affected in the hands.



Central Cord/Syrinx

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- A CSF containing cord cavity.
- Cervical region most commonly affected.
- Rarely in thoracic cord alone.
- Seen in 20-40% of Chiari I malformations.

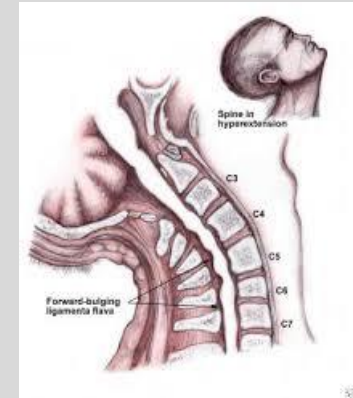
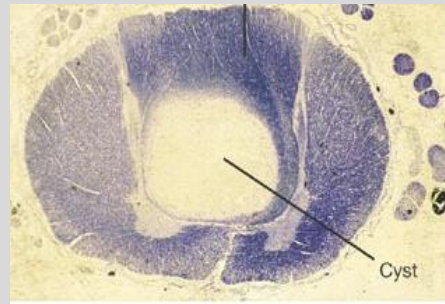
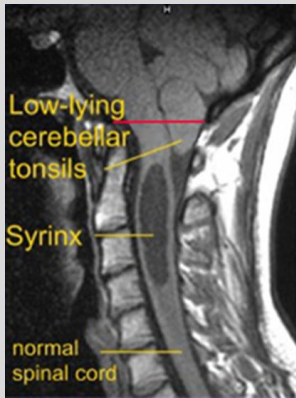
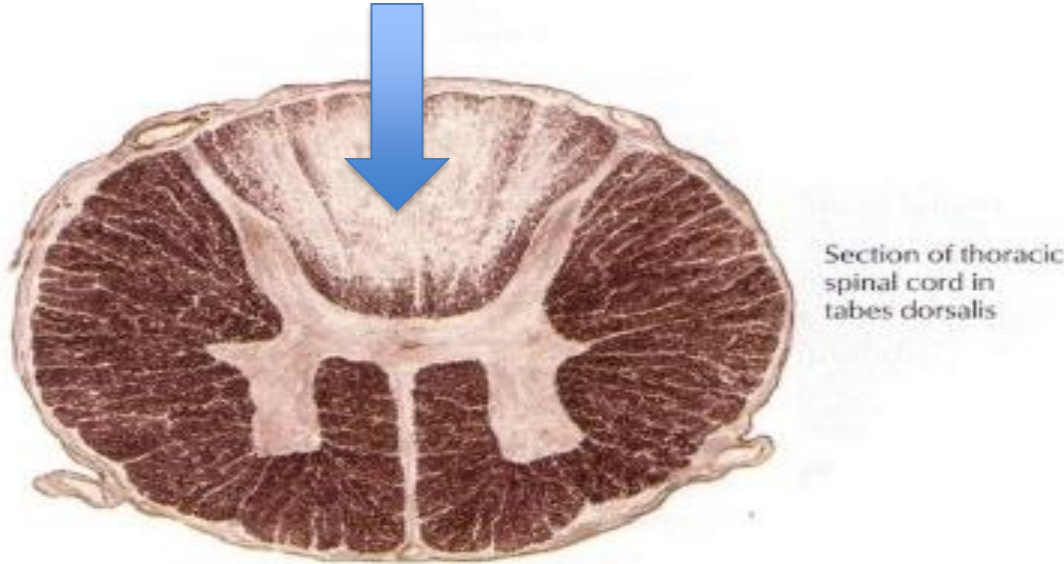


Image- emedicine.Medscape Localization in Clinical
Neurology. Wolters Kluwer. eighth edition. 2022

Dorsal Column Disorder

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http://library.med.utah.edu/neurologicexam/html/gait_abnormal.html#14

Etiology of Dorsal Column Syndrome

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- Tabes Dorsalis-syphilitic spinal cord disease
- Cervical spondylosis
- Posterior spinal artery infarction
- Multiple Sclerosis
- Partial cord trauma
- Post platinum-based chemo



Image: Ooscskills.com/ American Academy of Neurology- American Academy of Neurology- Continuum Spinal Cord Disorders, February, 2024 VOL 30, No 1

Tabes Dorsalis-Neurosyphilis

- Caused by the spirochete *T. pallidum*.
- Occurs decades after primary infection (most common after 20 years).
- Progressive gait abnormalities-slapping
- Positive Romberg- swaying with the eyes closed

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Image: Oskill.com American Academy of Neurology- Continuum series- Spinal Cord Disorders, February, 2024 VOL 30, No 1

Lhermitte Sign



Image: Youtube.com

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Case 3

- A 58-year-old woman underwent bariatric surgery several months ago. She was non-compliant in following up with her surgeon.
- Today she presents with numbness and weakness of her feet.

Posterolateral Column

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- Involvement of posterior column and lateral corticospinal tracts.



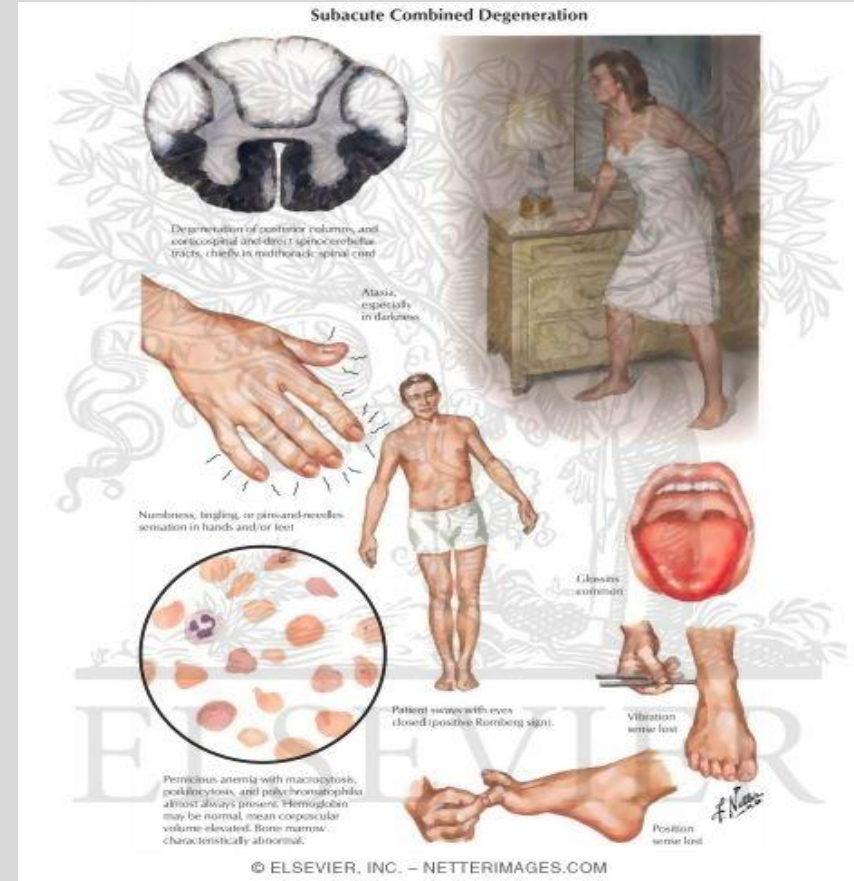
Netter plate

Subacute combined degen

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- Numbness
- Positive Romberg sign
- Glossitis
- Pernicious anemia
- Vibration and position sense lost

Netter plates- subacute combined degeneration

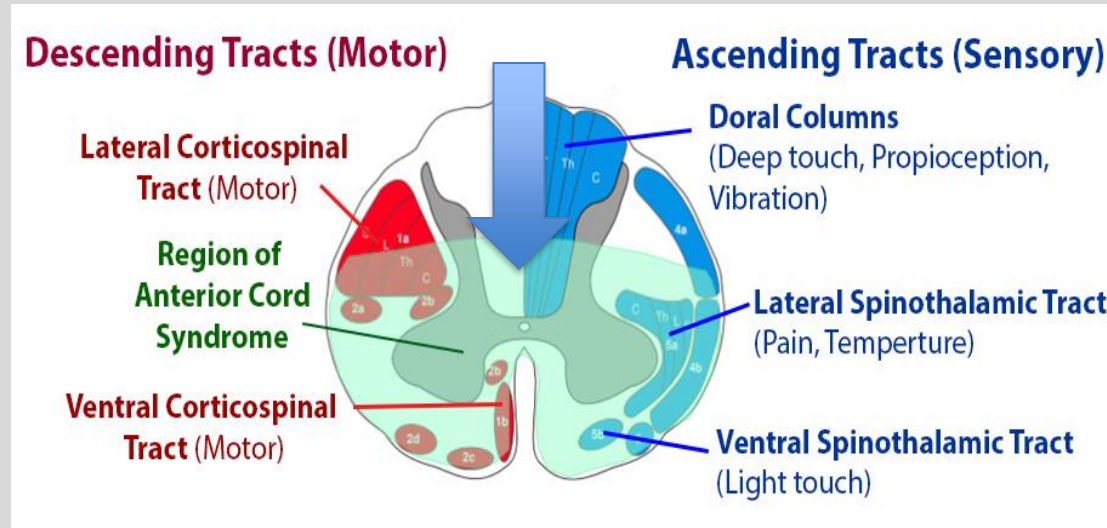


Anterior Cord Syndrome

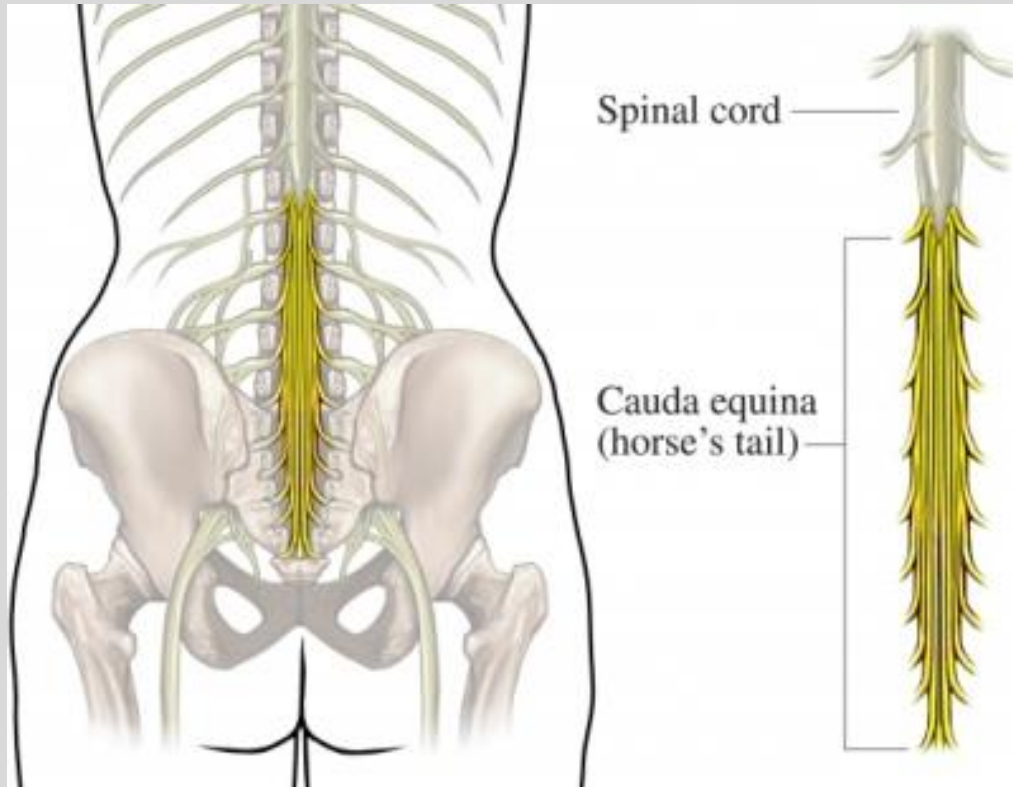
- Weakness and numbness
- Proprioception is spared

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Cauda Equina



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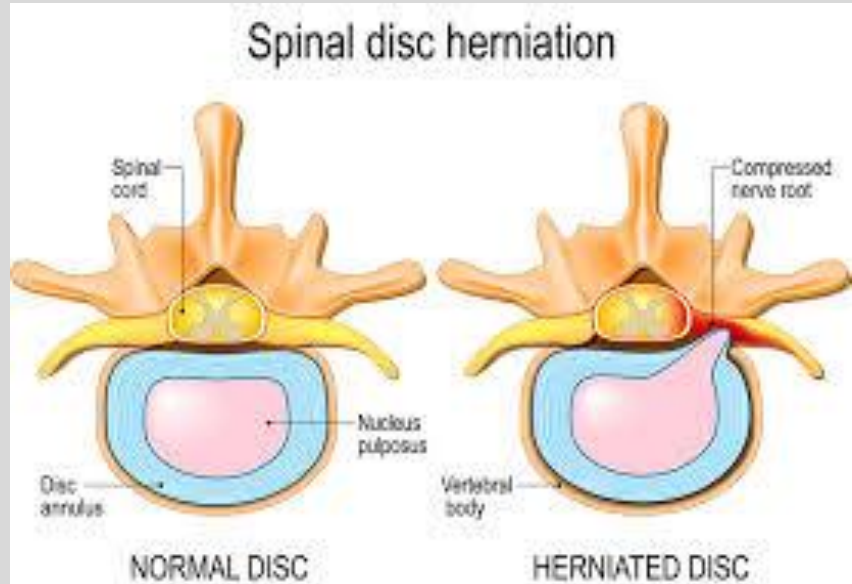
Premier surgery center of Louisville-cauda equina syndrome

Causes of Cauda Equina

- Intervertebral disc herniation (MC)
- Spinal fracture
- Iatrogenic cause
- Tumor

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Clinical Features CE

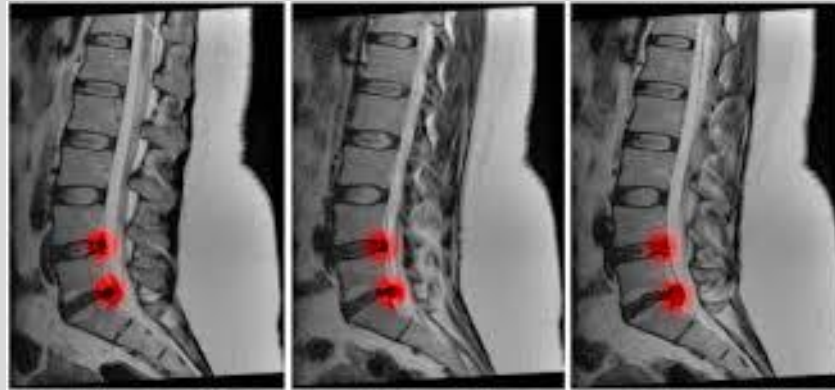
- Low back pain and radicular symptoms
- Bladder symptoms
- Bowel symptoms (less than urinary symptoms- constipation, rectal fullness)
- Sexual dysfunction
- Sensory loss- buttocks, posterior thighs, perineal region- MUST ASK!
- Weakness- **asymmetric**
- Reflexes- Reduced at knees and ankles.

Assess and Manage CE

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- MRI is the imaging modality of choice.
- Treatment is surgical decompression.
- Only occurs in 1-2% of cases.



Case 6

- A 49-year-old man presents to the ED after a bar fight. It resulted in a knife to his spine on the right.
- On examination there is weakness on the right side with loss of pain and temperature on the left side below the level of the lesion.

Brown Sequard Syndrome

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- Hemicord injury
- Disrupts descending ***already crossed*** corticospinal fibers
 - Segmental muscle weakness and atrophy and sensory signs.
 - Ipsilateral motor weakness
 - Ipsilateral vibratory/proprioceptive loss
 - Contralateral pain/temperature below the lesion.
- ***Often incomplete***

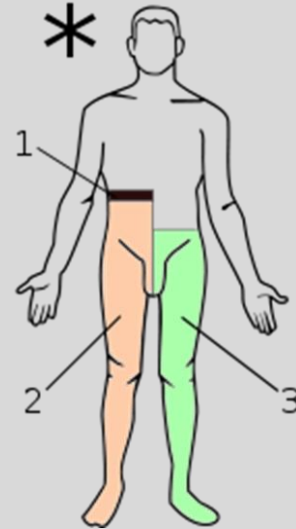


Image: quora.com Localization in Clinical Neurology, Wolters Kluwer, eighth edition, 2022

Review of cord lesions

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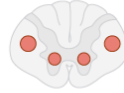
Spinal cord lesions

AREA AFFECTED

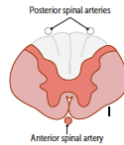
DISEASE



Spinal muscular atrophy



Amyotrophic lateral sclerosis



Complete occlusion of anterior
spinal artery



Tabes dorsalis



Syringomyelia



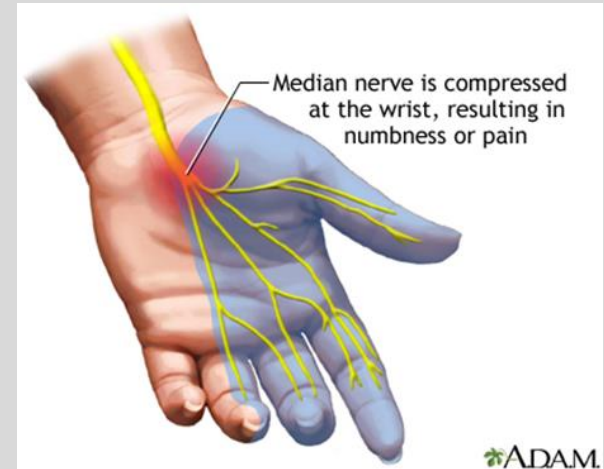
Vitamin B₁₂ deficiency

Median Neuropathy

- Carpal tunnel syndrome
- Wrist pain
- “Hand numbness”-sensory loss in the palmar aspect of the first 3 or 4 digits.
- Thenar weakness late in the course of the condition.

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Ulnar neuropathy

- Numbness in ½ 4th and 5th digit
- Clawing of 4th and 5th digit
- Weakness and wasting of the first dorsal interosseous muscle
- Repeated elbow trauma
- Finger Adductor weakness

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Radial Neuropathy

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- Unilateral painless wrist drop
- Finger or wrist extension weakness
- Painless gradually progressive bilateral wrist drop- lead poisoning.
- Alcoholic with Saturday night palsy (spiral groove is localization)

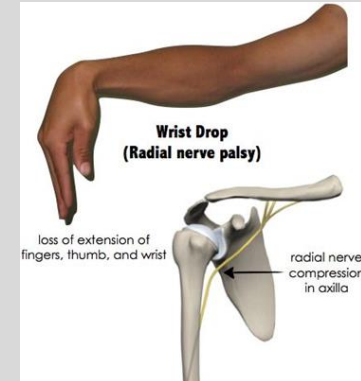
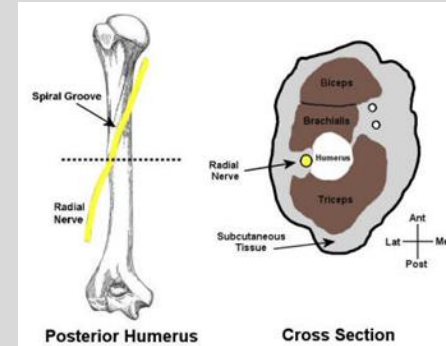
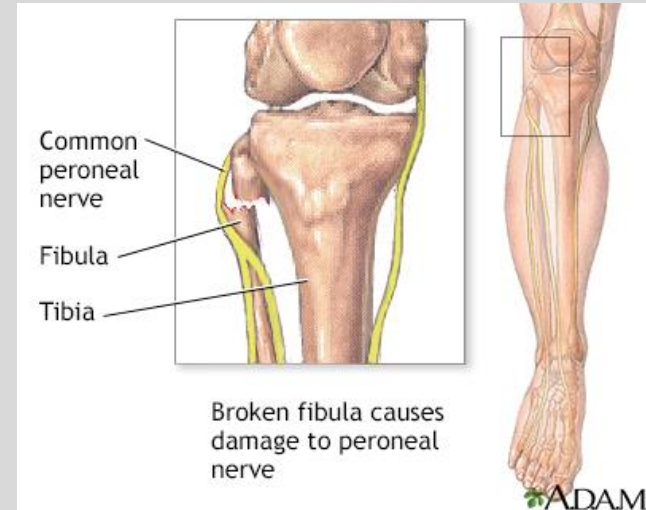


Image:school of medicine- case western/pinterest
Instant neurological diagnosis- Hawkes, Sethi, Swift

Fibular (peroneal neuropathy)

- Most frequent mononeuropathy in the lower extremity.
- Weakness of ankle dorsiflexion with foot drop may be seen.
- Trauma is the most common cause at the fibular head.
- May be seen with recent weight loss.



Meralgia Paresthetica

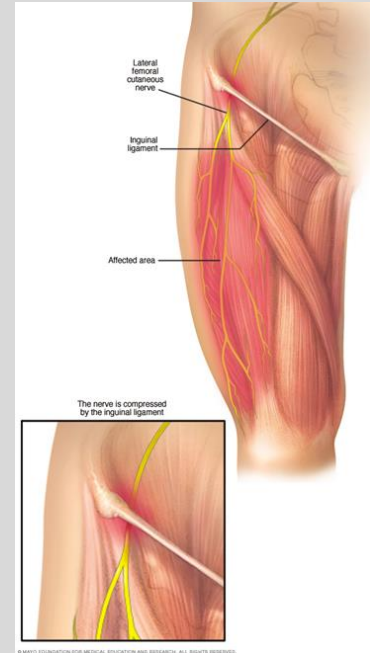
- Compression of the lateral femoral cutaneous nerve where it crosses under the inguinal ligament.
- Numbness of the lateral thigh.
- Pregnancy, weight gain, pelvic surgery

Image: mayo clinic

Instant neurological diagnosis-Hawkwa, Sethi, Swift

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